

TESTING ON LIVE APPLICATION

Theme: Online Learning Platform

Section A: Scenario Based Conceptual Questions

1. An online learning platform releases new courses every month. Identify the most suitable software development life cycle model and justify your choice.

Ans: The most suitable SDLC model is the Agile Model (Scrum).

Justification:

- Online learning platforms frequently add new courses and features.
- Requirements may change based on learner feedback and market trends.
- Agile supports incremental and iterative development.
- New features can be developed and released in short sprints (usually 2–4 weeks).
- Continuous testing and feedback help improve product quality.
- Agile enables faster delivery and quick adaptation to changes.
- Collaboration among developers, testers, and stakeholders is easier.

2. For a course purchase feature, define testing objectives from a quality assurance perspective.

Ans: The Main testing objectives are:

- Verify that users can successfully browse and purchase courses.
- Ensure accurate course pricing, discounts, and tax calculations.
- Validate payment processing through different payment methods.
- Confirm secure handling of user and payment information.

- Check that purchase confirmation emails and notifications are sent correctly.
- Ensure purchased courses are instantly accessible after successful payment.
- Verify error handling for failed or cancelled transactions.
- Confirm compatibility across different devices and browsers.
- Validate data accuracy in order history and invoices.
- Ensure the feature meets business and user requirements.

3. Explain how non-functional testing contributes to better learner experience.

Ans: Non-functional testing evaluates how well the system performs rather than what it does.

Contributions to learner experience:

- **Performance Testing:** Ensures fast page loading and smooth video streaming.
- **Load Testing:** Verifies the platform can handle many learners during peak hours.
- **Security Testing:** Protects personal information and payment details.
- **Usability Testing:** Ensures easy navigation and user-friendly interfaces.
- **Compatibility Testing:** Confirms the platform works on different devices, browsers, and operating systems.
- **Accessibility Testing:** Helps learners with disabilities access the platform effectively.
- **Reliability Testing:** Ensures minimal downtime and uninterrupted learning.

Effective non-functional testing improves user satisfaction, engagement, and trust.

4. During sprint execution, how should testers ensure acceptance criteria are testable?

Ans: testers should participate actively throughout the sprint.

Best practices:

- Review user stories during sprint planning.
- Ensure acceptance criteria are clear, specific, and measurable.
- Use the SMART principle (Specific, Measurable, Achievable, Relevant, Time-bound).
- Remove ambiguous terms such as "fast," "easy," or "user-friendly."
- Convert acceptance criteria into test scenarios and test cases.
- Collaborate with product owners and developers to clarify requirements.
- Use examples and business rules to define expected behaviour.
- Apply the Given–When–Then format for better understanding.
- Verify that acceptance criteria cover both positive and negative scenarios.

Example:

Given a user selects a course,

When the payment is completed successfully,

Then the course should appear in the user's dashboard immediately.

This approach ensures acceptance criteria are clear and testable.

5. A payment failure occurs after deployment. Assign severity and priority with justification.

Ans:

Severity: High / Critical

Priority: High

Justification:

- Payment functionality is a core business feature of an online learning platform.
- Users cannot purchase courses if payments fail.
- The issue directly impacts revenue generation.
- It negatively affects customer trust and user experience.
- Failed transactions may lead to customer complaints and loss of potential learners.
- Since the defect is found in the production environment, immediate action is required.

Therefore, the issue should be classified as:

Parameter	Level	Reason
Severity	High/Critical	Core functionality is broken, preventing course purchases.
Priority	High	Requires immediate resolution due to direct business impact.

Section B: Practical Manual Testing Tasks

1. STLC (Student Login): Document the full lifecycle including Requirement Analysis, Planning, Test Design, Execution, and Closure. Define specific Entry and Exit criteria for the Student Login module.

Ans: 1. STLC (Student Login Module)

Module: Student Login

Phase 1: Requirement Analysis

Activities:

- Understand login requirements and business rules.
- Identify input fields: Email/Username and Password.
- Analyze functional and non-functional requirements.
- Identify supported browsers and devices.
- Clarify password policies and session timeout requirements.

Requirements:

- Registered students should log in successfully using valid credentials.
- Invalid credentials should display an error message.
- Password field should mask entered characters.
- "Forgot Password" functionality should work correctly.
- Session should expire after a predefined inactive period.

Phase 2: Test Planning

Activities:

- Define testing scope and objectives.
- Estimate testing effort and schedule.
- Identify testing resources and responsibilities.

- Select testing techniques and tools.
- Define entry and exit criteria.

Test Types:

- Functional Testing
- UI Testing
- Security Testing
- Compatibility Testing
- Usability Testing
- Regression Testing

Phase 3: Test Design**Activities:**

- Create test scenarios and test cases.
- Prepare test data.
- Review and update test cases.
- Map requirements with test cases using RTM.

Sample Test Scenarios:

- Verify login with valid credentials.
- Verify login with invalid credentials.
- Verify mandatory field validations.
- Verify password masking.
- Verify "Forgot Password" functionality.
- Verify session timeout.
- Verify login on different browsers and devices.

Phase 4: Test Execution

Activities:

- Execute test cases.
- Record actual results.
- Compare expected and actual results.
- Log defects in the defect tracking tool.
- Retest fixed defects.
- Perform regression testing.

Example Execution Status:

TEST – CASE ID	FUNCTION NO	TEST SCENARIO	PRE- CONDITION	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	1	Valid Login	Account available	Valid credentials	Login successful	Dashboard displayed	Pass
2	2	Invalid Password	Account available	Invalid password	Error message displayed	Login denied	Pass
3	3	Empty Fields	Login page open	Blank fields	Validation message shown	Validation displayed	Pass
4	4	Forgot Password	Registered email	Valid email ID	Reset link sent	Reset email received	Pass
5	5	Session Timeout	User logged in	No activity	Auto logout	Session remained active	Fail

Phase 5: Test Closure

Activities:

- Verify all planned test cases are executed.
- Ensure critical defects are fixed.
- Prepare test summary report.
- Document lessons learned.
- Obtain stakeholder approval.
- Archive test artifacts.

Entry Criteria

- Requirements document approved.
- Login screen UI available.
- Test environment configured.
- Test accounts created.
- Test plan approved.
- Test data prepared.

Exit Criteria

- All planned test cases executed.
- 100% critical test cases passed.
- No open Critical or High severity defects.
- Test summary report completed.
- Stakeholder sign-off received.

2. Test Design (Course Rating 1-5): Apply Equivalence Partitioning and Boundary Value Analysis to the 1–5-star rating input. List the specific Valid and Invalid numeric test data required to verify the input limits.

Ans: Technique 1: Equivalence Partitioning (EP)

The valid rating range is **1 to 5**.

Valid Equivalence Class

- Values between **1 and 5**

Invalid Equivalence Classes

- Values less than **1**
- Values greater than **5**
- Blank or null values
- Non-numeric values

EP Test Data

Test Data	Class	Expected Result
3	Valid	Rating accepted
0	Invalid	Error message displayed
6	Invalid	Error message displayed
Blank	Invalid	Validation message displayed
A	Invalid	Error message displayed

Technique 2: Boundary Value Analysis (BVA)

Minimum Value = 1

Maximum Value = 5

BVA Test Data

Test Value	Type	Expected Result
0	Min - 1	Rejected
1	Min	Accepted
2	Min + 1	Accepted
4	Max - 1	Accepted
5	Max	Accepted
6	Max + 1	Rejected

Valid Numeric Test Data

1, 2, 3, 4, 5

Invalid Numeric Test Data

0, -1, 6, 7

3.Test Cases (Course Experience): Write one functional test case for each: Course Search & Browse, Course Learning Experience, and Assessments & Quizzes. Each case must include Steps, Expected Result, and Pass/Fail Status.

Ans:

T C I D	FUNCTI ON NO	TEST SCENARI O	PRE- CONDITI ON	TEST DATA	EXPECTE D RESULT	ACTUAL RESULT	STAT US
1	1	Course Search & Browse	User is logged in and courses are available	Search keyword: "Python"	Relevant courses matching the keyword should be displayed.	Relevant Python courses displayed successfully.	Pass
2	2	Course Learning Experience	User is logged in and enrolled in a course	Open lesson video	Course content should load, video controls should work, and progress should be saved.	Video played successfully and progress saved.	Pass

3	3	Assessments & Quizzes	User is logged in and quiz is available	Valid quiz answers	Quiz should be submitted successfully and score should be displayed.	Quiz submitted successfully and score displayed correctly.	Pass
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4. Defect Log: Document 3 unique defects found in the modules above. Each report must include: Defect Title, Severity, Steps to Reproduce, Actual vs. Expected Result, and Environment details.

Ans:

Defect ID	Defect Title	Severity	Steps to Reproduce	Expected Result	Actual Result	Environment
1	Search shows irrelevant courses	Medium	Search for "Python" courses.	Only relevant courses should appear.	Unrelated courses are displayed.	Windows 11, Chrome 137, UAT, Build v2.5.1
2	Video progress not saved	High	Play a video, refresh the page, reopen the lesson.	Video should resume from the last watched position.	Video starts from the beginning.	Windows 11, Firefox 139, Staging, Build v2.5.1

Section C: -Mini Capstone Project – End-to-End Testing Case Study Context

1. Test Planning Document: Prepare a Test Plan covering scope, objectives, testing types, environment, risks, assumptions, and entry and exit criteria

Ans:

1. Test Planning Document – Online Learning Platform

Project Name: Online Learning Platform – End-to-End Testing

Objective

To verify that the online learning platform functions correctly, securely, and efficiently from student login to course access after successful payment.

Scope

- Student registration and login
- Course search and selection
- Course purchase and payment
- Course access after purchase
- Assessments and quizzes
- Course ratings and reviews

Out of Scope

- Admin panel functionality
- Instructor course creation
- Third-party analytics integrations
- Mobile application testing (if not available)

Testing Objectives

- Validate all functional requirements.
- Verify secure payment processing.
- Ensure purchased courses are accessible immediately.
- Confirm system compatibility across browsers and devices.
- Identify defects before production release.

Testing Types

- Functional Testing
- Integration Testing
- System Testing
- Regression Testing
- Smoke Testing
- User Interface Testing
- Usability Testing
- Security Testing
- Performance Testing
- Compatibility Testing

Test Environment

Component	Details
Environment	UAT/Staging
Operating System	Windows 11, Android 15
Browsers	Chrome, Firefox, Edge
Database	MySQL
Build Version	v2.5.1

Network	Stable Internet Connection
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Assumptions

- Requirements are finalized and approved.
- Test environment is available.
- Test accounts and payment sandbox are configured.
- Third-party payment gateway is accessible.
- Test data is available.

Risks

- Third-party payment gateway downtime.
- Frequent requirement changes.
- Delayed defect fixes.
- Environment instability.
- Limited test data availability.

Entry Criteria

- Requirements document approved.
- Test environment configured.
- Test cases reviewed and approved.
- Test data prepared.
- Build deployed successfully.

Exit Criteria

- All planned test cases executed.
- 100% critical test cases passed.
- No open Critical or High severity defects.
- Test Summary Report prepared.
- Stakeholder approval received.

2. Requirements Traceability Matrix: Create a Requirements Traceability Matrix mapping requirements to test scenarios, test cases, and defects.

2. Requirements Traceability Matrix (RTM)

Requirement ID	Requirement Description	Test Scenario ID	Test Case ID	Defect ID	Test Status
REQ-001	Student should log in using valid credentials.	TS-001	TC-001	DEF-001	Pass
REQ-002	Student should search and browse courses.	TS-002	TC-002	NA	Pass
REQ-003	Student should select and purchase a course.	TS-003	TC-003	DEF-004	Pass
REQ-004	Payment should be processed successfully.	TS-004	TC-004	DEF-005	Pass
REQ-005	Purchased course should be accessible immediately.	TS-005	TC-005	DEF-006	Fail

REQ-006	Student should complete quizzes successfully.	TS-006	TC-006	DEF-003	Fail
REQ-007	Student should submit ratings between 1 and 5.	TS-007	TC-007	DEF-002	Fail

Coverage Summary

- Total Requirements: 7
- Requirements Covered: 7
- Test Cases Created: 7
- Defects Identified: 5
- Coverage Percentage: 100%

3.End-to-End Test Scenario: Write one detailed End-to-End test scenario covering: Login → Course Selection → Payment → Course Access

Ans:

3. End-to-End Test Scenario

Scenario ID

E2E-001

Scenario Name

Pre-Conditions

- Student account is registered and active.
- Course is available for purchase.
- Payment gateway is operational.
- Test payment credentials are available.

Test Steps

1. Open the Online Learning Platform.
2. Click the Login button.
3. Enter valid student credentials.
4. Click Sign In.
5. Search for a course using the search bar.
6. Select the desired course.
7. Review course details and pricing.
8. Click Enroll Now or Buy Now.
9. Choose a payment method.
10. Enter valid payment details.
11. Complete the payment process.
12. Verify payment confirmation message.

13. Navigate to My Courses.
14. Open the purchased course.
15. Access course videos, materials, and assessments.

Expected Result

- Student logs in successfully.
- Course search displays relevant results.
- Payment is processed successfully.
- Confirmation message and invoice are generated.
- Purchased course appears in My Courses.
- Student can access all course content without issues.

Actual Result

System behaves as expected.

Status

Pass

4. Test Summary Report: Prepare a Test Summary Report including execution status, defects identified, and release recommendation.

Ans:

Test Summary Report

Project Name: Online Learning Platform – Release v2.5.1

Testing Duration: 01 June 2026 – 07 June 2026

Test Execution Summary

Metric	Count
Total Test Cases	50
Executed Test Cases	50
Passed Test Cases	45
Failed Test Cases	5
Blocked Test Cases	0

Defect Summary

Severity	Count
Critical	1
High	1
Medium	2
Low	1

Key Defects Identified

- **DEF-001:** Login allowed access using expired passwords.
- **DEF-002:** Course rating accepted values greater than 5.
- **DEF-003:** Quiz results were not saved after submission.
- **DEF-004:** Payment failed intermittently during checkout.
- **DEF-006:** Purchased course was not visible immediately after payment.

Test Completion Status

- Functional Testing: Completed
- Regression Testing: Completed
- Integration Testing: Completed
- Security Testing: Completed
- UAT Testing: Completed

Release Recommendation

Conditional Go-Live

The application is recommended for release only after all Critical and High severity defects are fixed and successfully retested. Medium and Low severity defects can be scheduled for future releases based on business priority.

